

Securing Agentic AI Solutions

Agenda

- Need for Guardrails
- Security Risks - CIA Triad

Let's begin the discussion by answering a few questions.

Securing Agentic AI Solutions Quiz

Why are guardrails essential in LLM-based solutions?

A

To improve the model's training data.

B

To control and monitor inputs and outputs, ensuring safety and reliability.

C

To replace human supervision entirely.

D

To reduce model size.

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Guardrails in LLMs



Input Guardrail

Block harmful or unsafe prompts.

Stop users from tricking the model.

Allow only safe queries to pass to the LLM.

Output Guardrail

Filter unsafe or biased answers.

Ensure ethical and secure responses.

Keep outputs aligned with safety rules.

Securing Agentic AI Solutions Quiz

In a multi-agent AI system, an attacker inserts a malicious instruction that spreads from one agent to another and gradually gains higher privileges at each step. Which threat does this scenario best represent?

A

Cascading & Compounding Failures

B

Tool Misuse

C

Identity Abuse

D

Goal Hijacking

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Key Threats in Agentic Systems

Threat Type	Definition	How the Attack Happens	Potential Impact
Goal Hijacking	Manipulating an agent's objective through natural-language instructions.	An attacker injects prompts that redirect the agent away from its intended goal.	The agent performs unintended actions or generates misleading outcomes.
Cascading & Compounding Failures	Malicious behavior spreading across multiple agents and gaining authority at each stage.	A harmful payload passes through interconnected agents and accumulates additional permissions or influence.	The failure amplifies across the system, potentially causing large-scale disruption.
Tool Misuse	Exploiting the tools or APIs connected to an agent.	Attackers trick the agent into making unintended API calls, escalating privileges, or exploiting tool vulnerabilities.	Unauthorized actions, system compromise, or data manipulation.
Identity Abuse	Misuse of credentials that an agent uses on behalf of a user.	The agent accesses systems using valid user credentials but retrieves or exposes unauthorized enterprise data.	Sensitive information leakage or violation of access policies.
Communication Injection	Manipulating the communication channel between agents.	Attackers intercept or inject malicious messages into the inter-agent message bus.	Agents process false instructions, spreading malicious actions across workflows.



Power Ahead!

